Paleomagnetism and Potassium-Argon Ages of Some Volcanic Rocks from the Rio Grande Gorge, New Mexico[1]

M. OZIMA, M. KONO, I. KANEOKA, H. KINOSHITA, KAZUO KOBAYASHI, AND T. NAGATA

Geophysical Institute, University of Tokyo, Tokyo, Japan

E. E. LARSON[2] AND D. STRANGWAY

Department of Geology and Geophysics, Massachusetts Institute of Technology Cambridge, Massachusetts 02139

Successive lava flows exposed in the Rio Grande gorge, near Taos, New Mexico, can be divided into three groups, each of different magnetic polarity: the upper and lower groups have a normal polarity, whereas the middle one has a reversed polarity. K-Ar radiometric ages indicate that the base of the upper normal polarity sequence is about 3.7 m.y. old and that the reversed flows range from 3.7 m.y. to 4.4 m.y. The lowermost normal flow has the K-Ar age of 4.5 m.y. The age difference between the base of the upper normal and the top of the adjacent reversed flow group is about 0.05 m.y., indicating rapid polarity switching.

INTRODUCTION

It seems now to be well established that the geomagnetic field has undergone change in polarity during geological time. On the basis of K-Ar dating, *Doell et al.* [1966] showed that during the later Tertiary and Quaternary the geomagnetic field was comprised principally of successive normal and reversed polarity intervals, roughly of one million years each, with other more brief intervals of both polarities superimposed.

Only a few attempts to estimate the transition intervals from one polarity to the other have, however, been made. From their studies on Tertiary Iceland basalts, *Hospers* [1954] and *Sigurgeirsson* [1957] have estimated the transition interval as being of the order of 10^3–10^4 years. About 12,000 years has been suggested by *Van Zijl et al.* [1962] from a study of successive lava flows of Triassic-Jurassic age. These values, based on geological observation of the lava flow successions, may be subjected to large uncertainties. *Cox et al.* [1965], deducing from the overlaps of K-Ar ages of normal and reversed polarity sections in the late Tertiary and Quaternary lavas, suggested that the transition time should be much shorter than 50,000 years.

In view of the central importance of this problem to the theory of the origin of the geomagnetic field, it seems desirable to estimate the transition span by determination of the radiometric ages of lava flows that recorded a polarity switching. The present paper reports the results of paleomagnetic and K-Ar radiometric age measurements of some successive lava flows that recorded a polarity change from a reversed to a normal.

GEOLOGICAL DESCRIPTIONS OF SAMPLING SITES

West of Taos, New Mexico, Tertiary-Quaternary lava flows extend over approximately 500 km^2 (Figure 1). Evidently these lavas erupted from vents just west of the Rio Grande and covered this area in an essentially flat-lying succession. Interbedded with the flows are discontinuous layers of generally conglomeratic sediments which were derived from the Sangre de Cristo Mountains. *McKinlay* [1957] described the lava as olivine basalt. The basalts are well exposed in the Rio Grande gorge, and eleven lava flows were clearly identified in the cliff on the west side of the gorge at the bridge site (see Figure 1). At the Rio Taos site, which

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[2] Now at Department of Geology, University of Colorado, Boulder, Colorado 80302.

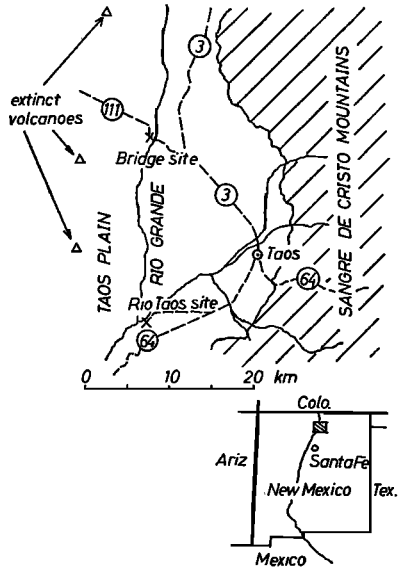


Fig. 1. Geological map of sampling sites.

is located about 20 km south of the bridge site (see Figure 1), three lava flows underlying the bridge site sequence were identified. Chemical analyses and petrological examination of each flow show that all the lavas have nearly identical chemical and mineralogic compositions and textural relationship, thereby suggesting a comagmatic origin (Aoki, private communication).

In total 111 core samples were obtained from 14 lava flows. The thickness of a flow ranges from 5 to 20 meters. It was at first suspected that for thick lava flows the time when the upper and lower parts of the flow reached the Curie point might differ from time for the central part. If considerable secular variation occurred during cooling, the direction of the remanent magnetization of the different flow zones could differ significantly. Hence, from the thicker flows, cores were taken at three different levels, that is, at the top, middle, and base. For thinner flows, samples were obtained at one level only.

Paleomagnetic Measurements

The directions of NRM's were measured with a spinner magnetometer. Samples from each lava flow were then subjected to (1) a storage test in which samples were kept for 3 to 6 months with their NRM in random direction to the geomagnetic field, and (2) low-temperature demagnetization in a nonmagnetic field at liquid nitrogen temperature [*Ozima et al.*, 1964]. Samples that showed change in their directions of NRM by more than 5° were rejected. After these stability tests, about half of the flows were further tested by ac and thermal demagnetization. Even after the demagnetization at the peak field of 200 oe and the thermal demagnetization at 250°C, directions of NRM's did not differ noticeably from the original ones. It is interesting to note that in the case of thicker flows, there were no observed significant differences between the direction of NRM's of samples taken at the bottom, top, and middle of a flow. Table 1 lists the results of the paleomagnetic measurements.

Figure 2 shows virtual poles for each flow. The directions of the normal and reversed NRM are nearly opposite, indicating good stability of the reversed NRM. The stability of the NRM's is further substantiated by the direction of the remanent magnetization of baked sediments adjacent to flows 829 and 820: a sediment baked by normal flow 829 shows normal polarity, and a baked sediment underlying reversed flow 820 is reversed.

Intensity of the geomagnetic field at the time of the formation of these lava flows has also been estimated, using Thellier's stepwise thermal demagnetization method [*Thellier and Thellier*, 1959] by two of the present authors [*Kono and Nagata*, 1966]. The result showed that the intensity of the geomagnetic field was nearly the same as that of the present one whether the polarity was normal or reversed.

K-Ar Radiometric Ages of Lavas

Choice of samples. K-Ar measurements were made on the cores used for the paleomagnetic study. As cores taken a few meters apart within the same flow sometimes showed considerable variation in K and Ar contents (see Table 2, core 826-15 and core 826-13B), both K and Ar analyses were made on each core specimen. Cores (diameter = 2.4 cm and height = 2.3 cm) were cut into four pieces; two pieces were used for Ar analysis and two for K analyses. Lavas are generally composed of 12 to 20% olivines, 0 to 20% pyroxenes,

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TABLE 1

Lava	N*	I,† deg	D,‡, deg	k§	α_{95}‖	Virtual Pole¶ Lat.	Virtual Pole¶ Long.	Polarity
				Bridge Site				
830	5	38.9	18.8	14.1	21.1	68.2	20.7	Normal
829	6	43.7	−10.4	9.7	22.7	75.9	116.4	Normal
827	5	−44.3	178.0	219	5.2	−79.4	−95.9	Reversed
826	9	−43.2	171.8	46.2	7.7	−76.7	−139.8	Reversed
825	9	−63.2	−145.6	12.8	15.0	−62.8	135.8	Reversed
824	8	−48.0	163.5	17.1	13.8	−74.3	−39.2	Reversed
823	8	−50.0	168.2	60.5	7.2	−78.7	−42.3	Reversed
822	9	−46.3	160.8	43.3	7.9	−71.5	−39.0	Reversed
820*	9	−49.7	164.7	329	2.8	−75.9	−36.6	Reversed
821	9	−48.3	169.4	322	2.9	−78.6	−51.7	Reversed
819	8	−6.0	135.5	2.3	57.3	...	...	Reversed
				Rio Taos Site				
833	2	−33.6	162.0	...	...	...	...	Reversed
832	9	−67.5	166.0	98.3	5.2	−72.8	42.9	Reversed
831	5	54.7	28.5	208	5.3	66.9	−21.5	Normal

* N, number of stable samples.
† I, inclination of mean direction of NRM.
‡ D, declination of mean direction of NRM.
§ k, Fisher's precision parameter.
‖ α_{95}, semiangle of cone of 95% confidence for mean direction.
¶ Virtual pole, positive for northern latitude/eastern longitude.

about 5% plagioclase, and 2 to 5% opaque minerals and others including a few per cent of volcanic glasses. For each lava flow, a thin section was prepared from a core drilled adjacent to the core used for the K-Ar dating. Samples showing any indication of alteration other than alteration of olivine were rejected. Further detailed petrographic description of lava flows is given in appendix.

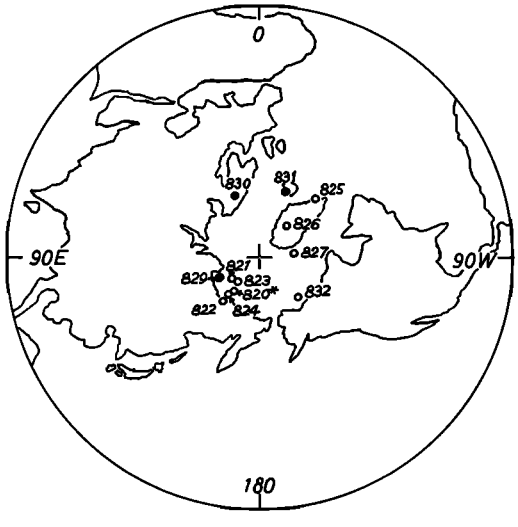


Fig. 2. Equal area projection of paleomagnetic virtual pole positions of lava flows. Solid circles represent normal polarity; open circles, reversed polarity.

Ar analysis. About 7–15 grams of total rock sample were used for Ar extraction. Samples were preheated overnight at 200°C in a high vacuum pyrex line.

Radiogenic Ar was extracted by heating samples at 1300°C for 1 hour and was then purified by CuO at 350°C and titanium sponge at 950°C. Quantitative determinations of radiogenic Ar^{40} were made by isotope dilution method, using a 99.99% pure Ar^{38} spike of about 5×10^{-6} cc STP by means of a 60°, 15-cm radius, Reynolds-type mass spectrometer. Isotopic analyses were made statically. In all cases the isotopic ratios Ar^{36}/Ar^{40} and Ar^{38}/Ar^{40} were determined by extrapolating the ratios to zero time, when the sample was introduced into the spectrometer. Mass discrimination was occasionally checked by analyzing purified air Ar. Ar^{36}/Ar^{40} was usually within 1% of the value given by Nier and no attempt

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TABLE 2

Sample	$(Ar^{40})_{rad}$, $\times 10^{-12}$ moles/g	$\frac{(Ar^{40})_{atm}}{(Ar^{40})_{total}}$, %	K, %	$\frac{(Ar^{40})_{rad}}{K^{40}}$, $\times 10^{-4}$	Age,* m.y.
830–15	2.979	95.3	0.406, 0.386	2.472	4.22
–13	3.079	93.5	0.478, 0.487	2.123	3.62
829–13	3.332	93.2	0.508, 0.502	2.155	3.68
–12	3.151	93.8	0.483, 0.480†	2.151	3.67
827–15	1.131	95.9	0.271, 0.252	1.419	2.43
826–15	3.397	83.3	0.501, 0.523	2.172	3.70
–13B	3.555	86.7	0.534, 0.535	2.184	3.73
820–23	7.003	84.5	1.010, 0.970	2.373	4.05
–232	6.994	79.1	0.993†	2.369	4.04
821–11	3.248	93.8	0.422, 0.411	2.566	4.38
831–13	7.119	83.3	0.877, 0.870	2.666	4.56
–15	7.144	78.8	0.893, 0.887	2.629	4.49

* $\lambda_e = 0.585 \times 10^{-10}$ yr^{-1}; $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}; $K^{40}/K = 1.19 \times 10^{-4}$ mole/mole.
† Measured as potassium tetraphenylboron salt.

was made to correct for discrimination in the radiogenic Ar^{40} determination.

K analysis. Most K analyses were made by isotope dilution method using a 60°, 22-cm radius, single-filament mass spectrometer with an electron multiplier. After a specimen was crushed to −150 mesh, about 0.03–0.06 grams of material were dissolved by HF and $HClO_4$ and evaporated to dryness. The precipitate was dissolved by 2.5 *N* HCl and mounted on a tantalum filament for isotopic analysis. Mass discrimination was occasionally checked by analyzing some KCl shelf solution. When a $(m_1/m_2)^{1/2}$ correction was made, K^{39}/K^{41} ratios were usually found within 1% of the value given by Nier. Hence, all isotopic ratios measured were corrected for this discrimination by multiplying by $(m_1/m_2)^{1/2}$. In two cases K contents were determined as tetraphenylboron salt, and the results were in good agreement with those obtained by the isotope dilution method.

Estimation of errors in K and Ar analyses and in calculated ages. In most cases, K analyses were made in duplicate, and the measurements were reproducible to about 4%. The averaged values of the duplicate K analyses were then used to calculate ages. For Ar analyses, however, it is difficult to be sure that two different whole rock pieces from the same flow are identical with respect to Ar content. Hence, results of Ar analyses on different cores from the same flow do not necessarily give an estimation of reproducibility of Ar analyses. Results from samples 820–23 and 820–232, which were made from the same core, may give the measure of reproducibility, which is less than 1%. The precision of the experiments may, however, be better judged by comparing the ages obtained for different cores from the same flow.

It is difficult to estimate the absolute error in the age thus determined. Formal error estimation shows that a 1% analytical error in the isotopic determination of Ar^{36}/Ar^{40} necessarily results in an uncertainty of 9% in the determination of radiogenic Ar^{40} for 90% atmospheric Ar contamination [*Lipson*, 1958]. In general, atmospheric Ar seems to be tightly bound in the matrix of extrusive rocks, and it is difficult to reduce the air contamination by preheating samples without losing radiogenic Ar. This was particularly true in the case of the Rio Grande basalts which have very many small vesicles. In many cases atmospheric contamination amounts to more than 85% even after preheating samples overnight at 200°C. Apparently the age thus obtained may be expected to be subject to an uncertainty of 7%. The differences in the ages obtained for a single flow are, however, usually much less than 0.1 m.y. In addition, the ages obtained for each flow are generally in an excellent accordance with the succession order of the lava flows. On the basis of

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these internal consistencies in ages within a flow and among flows, we conclude that the analytical errors in the ages thus obtained are of the order of 0.1 m.y. or less.

Ages of lava flows. Table 2 gives the results of Ar and K analyses on each of seven flows. Calculated ages are given also in this table and in Figure 3. Samples from the same flow usually give ages of nearly the same value, and the ages for the different flows are generally in the excellent accordance with the succession order of the lava flows. Samples 830–15 and 827–15, however, give anomalous ages. Thin sections show that both samples are fresh: alteration of samples is therefore an unlikely explanation for the anomalous values. It is likely that high air contamination in the extracted argon in both samples caused large experimental uncertainty, giving the anomalous age values. In addition, sample 827 has rather significant amounts of glasses (about 15%), which are less abundant in other samples. This might also be responsible for the anomalous ages.

Discussion

Stability tests and the consistency of the remanent magnetization between the lava flows and adjacent baked sediments strongly indicate that the NRM's represent the paleomagnetic field. The results of the K-Ar radiometric dating are generally in good accordance with the polarity time scale proposed by *Doell et al.* [1966]: the top layers of normal polarity (flows 830 and 829) corresponding to the 'Gauss normal polarity' (2.4 to 3.35 m.y.) and the underlying flows of reversed polarity to the 'Gilbert reversed polarity epoch' (3.35 m.y. —). The present study indicates, however, the time of the beginning of the Gauss normal epoch as being 0.35 m.y. older than the age proposed by *Doell et al.* [1966]. Systematic errors in experimental procedures are the most likely explanation for the discrepancy. The lower limit for the Gilbert reversed polarity epoch (3.35 m.y. —), however, has not yet been established by *Doell et al.* [1966].

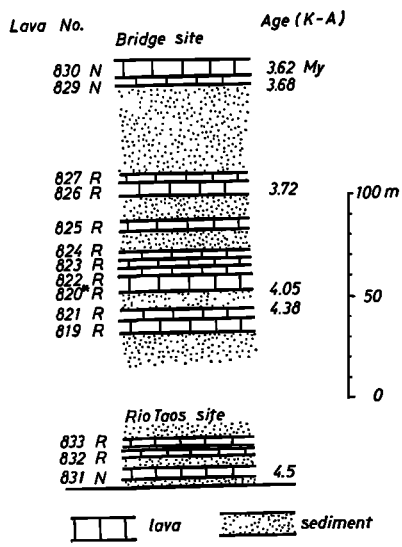


Fig. 3. Schematic representation of lava flow successions at the Bridge site and at the Rio Taos site. N and R indicate normal and reversed polarities of NRM's in each flow.

One must consider then whether the ages of the lowermost flow of normal polarity (flow 831) establish the lower limit of the Gilbert epoch and represents part of another normal polarity *epoch*, or whether they indicate a brief normal polarity *event* within the Gilbert *epoch*. Since volcanism is intermittent, a sequence of lava flows need not record the complete process of the geomagnetic polarity transition. The age difference between adjacent normal and reversed flows in a succession of lava flows must, however, set the upper limit to the time interval of the polarity switching. The age difference between the lowermost normal flow (829) and the underlying uppermost reversed flow (826) is about 0.05 m.y. As mentioned in the preceding section, a formal estimation of error involved in the determination of radiogenic Ar^{40} amounts to an error as high as 9% at 90% air Ar contamination level. The difference between ages obtained on a single flow is, however, usually less than 0.1 m.y., and it is reasonable to suppose that the face value of 0.05 m.y. would give a fairly good estimate of the maximum for the transition interval of this geomagnetic polarity switching. In this connection it is interesting that other investigators also estimated the interval as being of the order of 10^4 years. It remains to be seen whether such a rapid polarity switching as 0.05 m.y. was typical only for the 'Gauss-Gilbert' transition or is, rather, a general feature of all other transitions. Paleomagnetic and K-Ar radiometric age studies of other lava sequences that record the geomag-

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netic polarity change are in progress in our laboratories.

Conclusions

A succession of lava flows exposed in the Rio Grande gorge, near Taos, New Mexico, consists of three groups of lava flows of different magnetic polarities. The upper and lower groups have normal polarities, whereas the middle flows have a reversed polarity. Stability tests of the NRM's conclusively indicate that the polarities represent the paleomagnetic field.

K–Ar radiometric ages show that the upper normal polarity commenced about 3.7 m.y. ago, and the underlying reversed flows range from 3.7 to 4.4 m.y. The age of the lowermost flows of normal polarity is about 4.5 m.y. In comparison with a paleomagnetic polarity time scale proposed by *Doell et al.* [1966], it appears that the upper normal and the intermediate reversed groups correspond to the Gauss normal polarity epoch and Gilbert reversed polarity epoch, respectively. It is uncertain, however, whether the lowermost normal group truly represents another normal polarity *epoch* or indicates only a brief *event*.

The age difference between the lowermost flows of the upper normal group and the upper portion of the adjacent reversed group is about 0.05 m.y. This value may be taken as a reasonable estimate of the maximum value for the geomagnetic polarity switching, thereby indicating a very rapid transition of the earth's magnetic field.

Appendix. Petrographic Description of Lava Flows

Flow 830. Thickness of lava flow is about 5.0 meters, and the samples were drilled about 0.5 meter above the bottom of the lava flow. Olivine basalt. Except olivine, all minerals are very fresh. The olivine is slightly altered to iddingsite. Less than 10% glasses are present. No clays, xenoliths, or xenocrysts are observed.

Flow 829. Thickness of lava flow is about 5.0 meters, and the samples were drilled about 0.5 meter above the bottom of the lava flow. Olivine basalt. Except olivine, all minerals are very fresh. The olivine is slightly altered to iddingsite. Less than 10% glasses are present. No clays, xenoliths, or xenocrysts are observed.

Flow 827. Thickness of lava flow is about 3.8 meters, and the samples were drilled about 0.5 meter above the bottom of the lava flow. Olivine basalt. Except olivine, all other minerals are very fresh. The olivine is slightly altered to iddingsite. About 15% glasses are present, though no clays are observed. Large olivine crystals are present, which are likely to be xenolith.

Flow 826. Thickness of lava flow is about 6.0 meters, and the samples were drilled about 0.5 meter above the bottom. Olivine basalt. Except olivine, all other minerals are very fresh. The olivine is partly altered to iddingsite. Less than 10% glasses are present. No clays, xenoliths, or xenocrysts are observed.

Flow 820. Thickness of lava flow is about 6.5 meters, and the samples were drilled about 1.5 meters above the bottom. Olivine basalt. Except olivine, all other minerals are very fresh. The olivine is slightly altered to iddingsite. Less than 10% glasses are present. A few large olivine crystals are present, which are likely to be xenolith. No clays are observed.

Flow 821. Thickness of lava flow is about 4.0 meters, and the samples were drilled about 0.5 meter above the bottom. Olivine basalt. Except olivine, other minerals are very fresh. The olivine is slightly altered to iddingsite. Less than 10% glasses are present, though no clays are present. A few large olivine crystals are present, which are likely to be xenolith.

Flow 831. Thickness of lava flow is about 3 meters, and the samples were drilled about 1.5 meters above the bottom. Olivine basalt. Olivines are considerably altered to iddingsite, though other minerals are fresh. Less than 10% glasses are present. No clays, xenoliths, or xenocrysts are observed.

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